## School of Computing and Creative Technologies

# Assessment Specification

## Module Details

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| **Module Code** | UFCFAF-30-3 |
| **Module Title** | Development of Information Systems Project |
| **Module Leader** | Steve Battle |
| **Module Tutors** | Dilshan Jayatilake, Steve Battle, Ryan Fellows |
| **Year** | 2025-26 |
| **Task** | Portfolio |
| **Total number of assessments for this module** | 2 |
| **Weighting** | 70% |

## Dates

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| **Date issued to students** | 01/10/2025 |
| **Date to be returned to students** | 21/05/2026 |
| **Submission Date** | 23/04/2026 |
| **Submission Place** | Blackboard |
| **Submission Time** | 14:00 |
| **Submission Notes** | Field Board: 15/06/2026 |

## Feedback

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| **Feedback provision will be** | Formative feedback will be provided if each part of the portfolio is submitted in a timely manner. Summative feedback will be provided after the submission date. |

# Contents

# Section 1: Overview of Assessment

This assignment assesses the following module learning outcomes:

- Analyse software architecture using suitable models and methods.
- Develop information systems using the techniques of Software Engineering Management.
- Utilise Software Configuration Management tools.
- Apply software testing methods.

This is a **Group-Based** assignment worth **70%** of the overall mark for the module. Groups of one are allowed. Maximum group size is 5 members.

Your group has been assigned the responsibility of modelling the business architecture for a client as described in the case study provided. You are expected to use industry-standard methods, tools and techniques for documenting and communicating your work.

The assignment is described in more detail in section 2.
If you have any questions about this assignment, please contact your module leader.

# Section 2: Task Specification

Engaging in this assignment will enable you to enhance your technical expertise and explore emerging trends in the field of information system development. It also provides an opportunity to conduct a critical assessment of prevailing methods in this domain. Additionally, this assignment will help you nurture essential soft skills highly sought after by the industry, including effective communication, time management, and collaborative teamwork.

- You will first develop a **socio-technical** model of the client system using **i***. This will highlight the actors and activities involved, as well as (soft and hard) business goals.
- You have been tasked to construct **Strategic Business Process** model that add additional detail to each of the activities identified. This lays the foundation for creating operational business models that can be directly automated, with the addition of forms for capturing user input. Use the Business Process Modelling Notation (BPMN).
- You are also tasked to design an **Operational process model** based on the Strategic Business Process (Client requirements). Your models should expand beyond a single pool. An example of such BPMN process can be found in Additional Resources provided with this document. You must automate this process using a process automation engine such as Camunda.
- Your system should provide user-friendly interfaces for the users to interact with the system. You can use any technology that you deem suitable, though Camunda provides Camunda forms to capture user input. You must consider aspects of reusability, data validation, efficiency, data persistence and the effectiveness of your system.
- A test strategy must be developed, and the automation thoroughly tested.

# Section 3: Deliverables

| Item | Detail |
|---|---|
| Socio Technical Model | Develop a Socio-technical model of the client system using i*. You will model the information system you propose from the perspective of a socio-technical system using i* framework. You should develop a Strategic Dependency (SD) model and elaborated to Strategic Rationale (SR) Model.<br><br>**Deliverables**:<br>• Strategic Dependency (SD) model<br>• Strategic Rationale (SR) Model<br><br>**Format**: Link to the folder in your repository where you have stored the models. These modes must be stored in .pdf or docx with the .txt file of the model **(Check your tutors have access to these repositories)** |
| Business Process Modelling | Construct a strategic business process model to represent the client's AS-IS business process. Use BPMN 2.0 to create relevant diagrams.<br>Design and Automate a BPMN 2.0 operational model for the client. A fully automated business process with comprehensive functionalities as outlined in the provided case study. This should encompass external services/workers, demonstrate adherence to best practices in software engineering, and provide evidence of effective project management and work distribution.<br><br>**Deliverables**:<br>• Strategic business process model<br>• Operational Business Process Model<br>• Any other external services/UIs or any relevant artefacts of the project with a ReadMe if any configurations needed to run your project.<br><br>**Format**: Link to your repository or Camunda modeller **(Check your tutors have access to these repositories)** |
| Testing | You must provide clear evidence of your testing process, including the chosen test strategy, test cases, results, and any defects identified. The report should demonstrate how the model was validated and outline any limitations or areas requiring further refinement.<br><br>**Deliverables**: Testing Report<br>**Format**: Link to the folder in your repository where you have stored the testing evidence. These modes must be stored in .pdf or docx with the .txt file of the model **(Check your tutors have access to these repositories)** |
| Configuration Management | You are required to demonstrate evidence of applying configuration management principles during this assessment. This can be achieved by employing a configuration management tool of your choice. When submitting a link to your repository, ensure that it is accessible to the entire module team.<br><br>**Deliverables**: Repository<br>**Format**: Link to the repository where you have stored your project **(Check your tutors have access to these repositories)** |

# Section 4: Marking Criteria

| Criteria | < 40% | 40% - 59% | 60% - 69% | ≧ 70% |
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| **i* Socio-technical model** | Presents limited evidence of a sociotechnical model, or the model is weakly specified. | A complete sociotechnical model with Strategic Dependencies and Strategic Rationale. | A well-considered, sociotechnical model, including correct and appropriate identification of social elements including clearly defined hard and soft goals. | Shows originality and deep understanding of the issues relating to the case study, supported by rational justification. |
| **Strategic BPMN** | Limited evidence of a basic business process, or the business process is weakly specified. | A complete strategic business process, including correct human task definitions. A good description of the process. | A well-considered, business process, including correct and appropriate human-task definitions. | Shows exceptional BPM with originality and a deep understanding of issues relating to human tasks. |
| **Operational BPMN** | Presents limited evidence of an operational model, or the model is weakly specified. | A complete elaborate business process, including correct service tasks and pools. Good description of the process. | A well-considered elaborate business process, including correct and appropriate service tasks and pools. | Exceptional BPMN with originality and a deep understanding of issues related service tasks, pools, and sub-processes. |
| **Forms** | Limited or no evidence of using UIs to capture user inputs. | A comprehensive design of UI(s) to capture user inputs with no input validation. | A comprehensive design of UI(s) to capture user inputs with some input validation | Exhibits a comprehensive and exceptional design of UIs to capture user inputs, with thorough input validation and demonstrated consideration of UX. |
| **Automation** | The business process is either not automated or the automation is highly flawed and incomplete. | Business process is partially automated, but with significant gaps and flaws. Integration of external services or workers is minimal and poorly executed. | Business process is well-automated, with only minor gaps or flaws. Integration of external services or workers is effective and well-executed. Adherence to best practices. | Business process is fully automated with no significant gaps or flaws. Integration of external services or workers is seamless and highly effective. |
| **Testing** | No evidence of a test strategy, or the execution is highly flawed. | Limited execution of the test strategy, with significant gaps. Some parts of the test strategy may be implemented, but many are missing or poorly executed. | A good execution of the test strategy, with only minor gaps. Most elements are well-developed and coherent. The analysis is clear and thorough, identifying key issues and areas for improvement. | The test strategy is fully and effectively executed, showcasing a high level of understanding and application. The analysis is thorough and insightful, clearly identifying issues and areas for improvement. |
| **Evidence and application of configuration management tools and practices** | Little or no evidence of configuration management tools used; repository access is missing or disorganised. | Basic use of a configuration management tool demonstrated; repository is accessible but may be poorly structured or lack clear version control practices. | Good use of configuration management tools with clear structure, version control and collaborative practices evident in the repository. | Excellent use of configuration management tools demonstrating professional practices; well-documented and structured repository showing consistent collaboration and version tracking. |

Individual students may receive an adjusted group mark:

- Students who have contributed above & beyond and demonstrated practical and intellectual leadership throughout, may receive a positive mark adjustment. We are not rewarding the quantity of your work, but your abilities as a team-player.
- Students who have been simply passengers, have disengaged from the project, or who demonstrate a lower quality of contributions may receive a negative mark adjustment.

# Section 5: Feedback mechanisms

Formative feedback will be provided for each part of the portfolio if the work is submitted in a timely manner.

# Section 6: Appendices

## 6.1 Completing your assessment

**Where should I start?**
This is a group-based assessment, so it would be good to get to know your group well. This involves understanding the strengths and weaknesses of your team.

**What do I need to do to pass?**
The minimum overall mark to pass is 40% aggregated across all criteria following the marking scheme enclosed in this document.

**How do I achieve high marks in this assessment?**
to achieve 1st class or 2:1, your work should demonstrate exceptional BPM originality and a deep understanding of the problem domain, exceptional design of UI(s) to capture user inputs with thorough input validation, professionally structured report with state of the art, academically sound references and professional looking presentations.

**How does the learning and teaching relate to the assessment?**
The lectures & practical sessions of Term 1 and 2 are related to this assessment.

**What additional resources may help me complete this assessment?**
Please see additional supporting material that provides examples of operational business processes.

## What do I do if I am concerned about completing this assessment?

UWE Bristol offer a range of Assessment Support Options that you can explore through this link, and both Academic Support and Wellbeing Support are available.

For further information, please see the Academic Survival Guide.

### 6.2 Assessment Content

In line with UWE Bristol's Assessment Content Limit Policy (formerly the Word Count Policy), word count includes all text, including (but not limited to): the main body of text (including headings), all citations (both in and out of brackets), text boxes, tables and graphs, figures and diagrams, quotes, lists.

### 6.3 Assessment Offences

## How do I avoid an Assessment Offence on this module? [2]

Explain things in your own words, not those of a machine. Use the support above if you feel unable to submit your own work for this module.

UWE Bristol's UWE's Assessment Offences Policy requires that you submit work that is entirely your own and reflects your own learning. It is important to:

- Ensure that you reference all sources used, using the UWE Harvard system. Use the guidance available on UWE's Study Skills referencing pages.
- Avoid copying and pasting any work into this assessment, including your own previous assessments, work from other students or internet sources
- Develop your own style, arguments and wording. Avoid copying and changing individual words but keeping essentially the same sentences and/or structures from other sources
- Never give your work to others who may copy it
- If you are doing an individual assessment, develop your own work and preparation. Do not allow anyone to make amendments to your work (including proof-readers, who may highlight issues but not edit the work).

**When submitting your work, you will be required to confirm that the work is your own.** Text-matching software and other methods are routinely used to check submissions against other submissions to the university and internet sources. Details of what constitutes plagiarism and how to avoid it can be found on UWE's Study Skills pages about avoiding plagiarism.

### 6.4 Use of Generative AI (ChatGPT or similar)

Generative AI may be used to suggest talking points but please ensure that any text is in your own voice.

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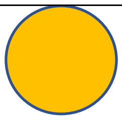
|  | You can use Generative AI to help with your coding, designing and articulating of your ideas but you must demonstrate a thorough understanding of whole project. |

### 6.5 Guidance on Referencing (inc AI):

Please note that the aim of referencing is to demonstrate you have read and understood a range of sources to evidence your key points. You need to list the references consistently and in such a way as to ensure the reader can follow up on the sources for themselves.
Referencing - Study skills | UWE Bristol
Using generative AI at UWE Bristol - Study skills | UWE Bristol

## 6.6 Studiosity

We recommend that you submit your assignment draft to the Studiosity writing feedback service before submitting it finally to Blackboard.

Studiosity will provide advice on how to improve your writing, focusing on your structure, use of language, spelling and grammar, use of sources and critical thinking.

You can submit your work as many times as you like - up to 8,000 words each time.

Please note, however, that it will not comment on the subject content of your assignment nor indicate what kind of mark you may receive.

Please see the student facing Studiosity webpage for more detail about the service on offer (including a Video walkthrough).